

GradeSense - Candidate Answer Sheet

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Total marks: 15

Answer 1 - Science

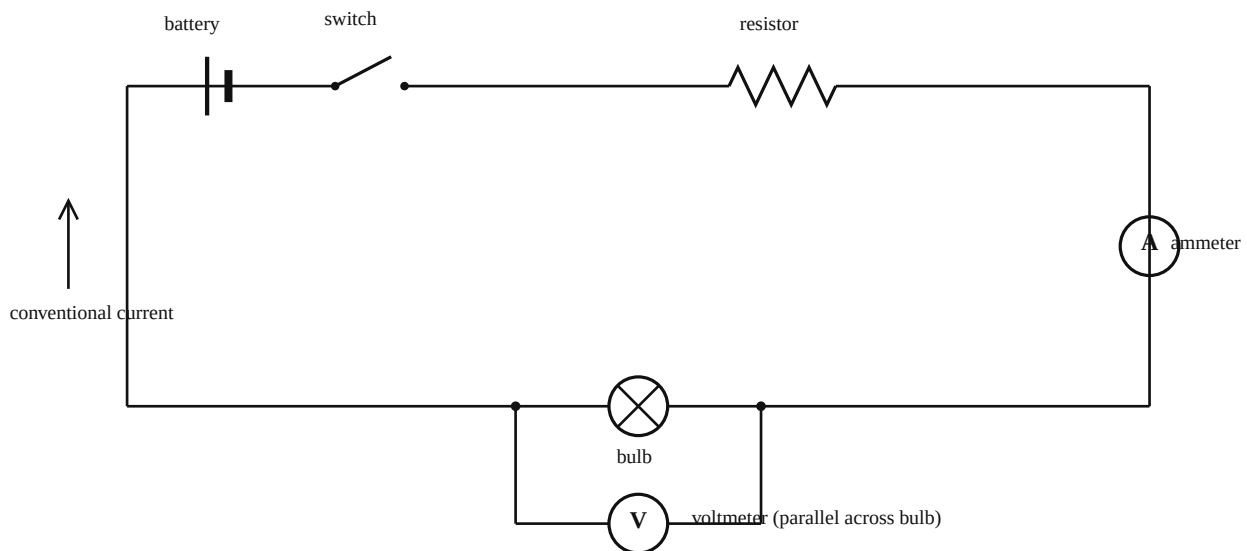
A simple electric circuit is a closed conducting path along which electric current can flow. The battery supplies the potential difference that drives the current around the circuit. The switch is used to open or close that path. When the switch is closed the path is complete and current flows through the resistor and the bulb; when the switch is open the path is broken and no current flows.

The battery, switch, resistor, bulb and ammeter are all connected in series in the main circuit, so the same current passes through each of them. The ammeter must be in series because it measures the current flowing through the circuit. The voltmeter is connected in parallel across the bulb, because it measures the potential difference between the two ends of the bulb.

In the diagram the conventional current direction is marked with an arrow, flowing out of the positive terminal of the battery, through the external circuit, and back into the negative terminal.

The current depends on both the potential difference and the resistance. By Ohm's law, $V = IR$. If the battery voltage stays constant and the resistance is increased, the current flowing through the circuit decreases. If the resistance is reduced, a larger current flows.

Circuit diagram



Answer 2 - English

In my view technology does not automatically make students better or worse learners; what matters is how the student chooses to use it. Access to information has clearly improved, but access on its own is not the same thing as understanding.

Technology genuinely helps when it is used to build understanding. A student who cannot follow a particular explanation of refraction in class can watch three different explanations at home and keep the one that finally makes sense. Digital libraries and interactive simulations also let a student work at their own pace, which a single classroom cannot do.

However, easy access creates a real risk of dependence. If a student looks up the solution to every difficult question immediately, the work gets finished but the reasoning is never practised, and the ability to attack an unfamiliar problem never develops. There is also the danger of accepting inaccurate information simply because it appeared online and looked confident.

I therefore conclude that technology should be treated as a tool that supports thinking rather than a replacement for it. Used to check and extend your own reasoning it makes you a stronger learner; used to skip the reasoning entirely it makes you a weaker one.

Answer 3 - Economics

The data is plotted with quantity on the horizontal axis and price on the vertical axis. The demand curve slopes downward from left to right, because consumers demand a greater quantity when the price is lower. The supply curve slopes upward from left to right, because producers are willing to supply a greater quantity when the price is higher.

The two curves intersect at a price of Rs 30 and a quantity of 60 units. This is the market equilibrium, because at that price the quantity demanded is exactly equal to the quantity supplied and there is no tendency for the price to move.

If the market price is below Rs 30, quantity demanded exceeds quantity supplied and there is a shortage: buyers want more than producers will sell, which pushes the price up. If the market price is above Rs 30, quantity supplied exceeds quantity demanded and there is a surplus: producers hold unsold goods, which pushes the price down.

If the cost of production rises, supplying each unit becomes less profitable at the existing price, so producers offer less at every price and the supply curve shifts to the left. The new supply curve meets the unchanged demand curve at a higher price and a lower quantity, so the new equilibrium has a higher equilibrium price and a lower equilibrium quantity. This shift is shown as the dashed curve on my graph.

Demand and supply graph

